

SAISH TIWARI

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PROFESSIONAL SUMMARY

Software Engineering student and AWS Certified Solutions Architect - Associate with hands-on experience building Java and Spring Boot backend applications, REST APIs, SQL databases, and cloud infrastructure. Experienced in designing reliable services, optimizing PostgreSQL/MySQL queries, containerizing applications with Docker, and deploying and monitoring workloads on AWS. Brings complementary exposure to data engineering, automation, and AI-assisted development, supported by a strong foundation in distributed systems, data validation, and scalable software design.

TECHNICAL SKILLS

Data & Analytics: SQL (PostgreSQL, MySQL), Python, R, Data Cleaning & Validation, 3NF Relational Modeling, ETL/ELT Pipelines, Source-to-Target Mapping, Data Dictionaries, Power BI, BI Reporting

Cloud & Backend: AWS (S3, RDS, Lambda, EC2, CloudWatch, IAM, VPC), REST APIs, Java, Spring Boot, JPA/Hibernate, Docker, Kubernetes

Tools & Foundations: Git, GitHub Actions, CI/CD, Bash, Quantitative Analysis, KPI Tracking, Data Integrity Audits, Agile Development

EDUCATION & CERTIFICATIONS

Bachelor of Science in Software Engineering | University of Bedfordshire

Currently Enrolled (Degree-Seeking)

Expected Graduation:

Jan 2027

AWS Certified Solutions Architect - Associate - Amazon Web Services

Active Certification

Cisco CCNA v7: Introduction to Networks

JPMorgan Chase Software Engineering Job Simulation - Forage

TECHNICAL EXPERIENCE

Java Backend Developer | Yaksha Soft

July 2025 - February 2026

- Developed and maintained high-throughput Spring Boot REST APIs that processed and validated relational data across PostgreSQL and MySQL databases.
- Optimized complex SQL queries, database indexing, and connection pools, significantly reducing data retrieval latency and improving system performance.
- Implemented strict input validation layers and exception-handling frameworks to safeguard transactional integrity and eliminate ingestion anomalies.
- Containerized backend services with Docker and supported GitHub Actions CI/CD workflows for seamless integration and deployment.

Cloud Engineering Apprentice | Adex International

April 2025 - June 2025

- Provisioned scalable AWS infrastructure utilizing S3, Lambda, RDS, and EC2 to support secure, automated data storage and cloud processing workloads.
- Configured IAM roles, VPC subnets, and security groups to ensure least-privilege access and enterprise data security standards.
- Built centralized CloudWatch monitoring dashboards, custom metrics, and alarms to track system health, operational KPIs, and pipeline reliability.
- Applied S3 lifecycle management policies and automated resource scripts to optimize cloud storage capacity and operational costs.

PROJECTS & DATA ENGINEERING SOLUTIONS

Regional Socioeconomic Data Pipeline & Analytics

Technologies: SQL, PostgreSQL, Python, R, Power BI, 3NF Data Modeling

- Developing an automated, reproducible ETL pipeline to extract, ingest, clean, and validate public socioeconomic datasets across disparate regional sources.
- Authored comprehensive data dictionaries and source-to-target mapping documentation to standardize schemas, field definitions, and KPI calculations.
- Architected a normalized PostgreSQL database (3NF) to support efficient analytical queries across demographics, labor, and economic observations.
- Created automated SQL validation scripts to audit schema consistency, identify missing values, and flag anomalies, ensuring 100% data accuracy.
- Built interactive reports and visualizations in R and Power BI to communicate key metric trends, variance, and comparative regional benchmarks.

Phoenix - Self-Healing Distributed Application Framework

Technologies: Java, Spring Boot, Maven, JUnit, Docker, YAML/JSON

- Built a Java-based framework for service registration and discovery, heartbeat telemetry, and active health checks across distributed applications.
- Designed a configuration-driven recovery engine that classifies failures and applies YAML/JSON policies for retries with exponential backoff and circuit-breaker protection.
- Centralized structured health and recovery logs into a unified event timeline, improving reliability monitoring and root-cause analysis across services.
- Validated recovery behavior through failure injection and JUnit scenarios; packaged optional Spring Boot sample services with Maven and Docker for repeatable testing.